

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
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Code of Conduct

I V.Rakshana(192425137) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V.Rakshana

Problem Statement 24: Smart Museum Artifact Preservation Platform

Abstract

Museums and cultural-heritage institutions hold irreplaceable artefacts that are continuously at risk from environmental degradation (temperature, humidity, light, pollutants), physical damage, theft, and loss of provenance information. A Smart Museum Artifact Preservation Platform aims to combine Internet of Things (IoT) sensing, computer-vision-based condition monitoring, and distributed-ledger technologies into a unified system for preventive conservation, damage assessment, and authenticity management. This report presents a literature survey of research relevant to the assigned problem statement. Twelve research works spanning IoT-based microclimate monitoring, deep-learning-based damage/crack detection, RFID-based anti-theft systems, and blockchain/digital-twin-based provenance platforms are reviewed.

1. Introduction

Cultural heritage artefacts housed in museums are exposed to multiple, simultaneous risks: uncontrolled temperature and relative-humidity fluctuations that cause cracking, warping, corrosion and mould growth; excessive light exposure that fades pigments and organic materials; airborne pollutants; physical damage from handling or ageing; and theft or unauthorised alteration of provenance records. Traditional preservation practice depends on periodic manual inspection using handheld instruments, which is labour-intensive, infrequent, and often fails to detect gradual or hidden degradation until damage is already visible. Recent advances in the Internet of Things, computer vision, deep learning, and blockchain technology have opened new possibilities for continuous, automated, and tamper-resistant artefact preservation.

2. Individual Paper Reviews

2.1 IoT-Based Environmental Monitoring Systems

2.1.1 Prasanth Ch.B.V. et al. (2021) — "IoT Based Environment Monitoring System To Protect Heritage Artefacts," IEEE

Objective: To continuously monitor and regulate the climatic conditions. Methodology/Technology: A microcontroller-based IoT node interfaced with light, temperature and humidity sensors, paired with an Android application for remote monitoring and light-intensity control. Findings: The system replaces traditional periodic manual measurement with continuous, automated sensing, allowing museum staff to intervene before adverse conditions damage the artefacts.

2.1.2 Eltresy et al. (2019) — "RF Energy Harvesting IoT System for Museum Ambience Control with Deep Learning," Sensors (MDPI)

Objective: To build a self-powered IoT monitoring and control system that also detects vandalism attempts. Methodology/Technology: Rectenna arrays harvest ambient RF energy to power always-on sensor nodes; a deep-learning model analyses collected environmental time-series data to identify daily trends and improve resilience to faulty sensor readings. Findings: The system autonomously optimises museum ambience, provides vandalism alarms, and demonstrates that battery-free operation is feasible for continuous museum sensing, although the amount of harvestable RF energy constrains sensor density.

2.1.3 Gawade et al. (2021) — "A Smart Archive Box for Museum Artifact Monitoring Using Battery-Less Temperature and Humidity Sensing," Sensors (MDPI)

Objective: To monitor the often-unknown microclimate inside enclosed storage boxes, which can differ significantly from the surrounding room environment. Methodology/Technology: A battery-less, Near-

Field-Communication-powered temperature/humidity sensor is embedded in a cardboard archive box and read using a standard smartphone. Findings: The approach enables non-invasive, long-term microclimate logging without opening the box, supporting preventive conservation, though it requires an operator to actively scan the box rather than providing continuous automated alerts.

2.2 Deep-Learning-Based Damage and Condition Assessment

2.2.1 Wang et al. (2018–2020 series) — Damage Classification and Segmentation for Historic Masonry Structures

Objective: To automate the detection and classification of physical damage (cracks, spalling, glazed-tile defects) in historic structures. Methodology/Technology: A progression of convolutional neural network (CNN) approaches — image classification, Faster R-CNN object detection, and Mask R-CNN instance segmentation — applied to still images and mobile-captured photographs. Findings: The models achieve near-expert-level detection accuracy and were successfully deployed on mobile devices for field inspection, though they require large labelled datasets and their accuracy is sensitive to lighting and material variability.

2.2.2 Applied Sciences (2025) — "Deep Learning-Based Crack Detection on Cultural Heritage Surfaces"

Objective: To precisely measure crack length on heritage surfaces without manual callipers. Methodology/Technology: A two-stage pipeline combining GoogleNet-based image classification with SegNet semantic segmentation, calibrated using a parallel-laser measurement technique. Findings: Crack-length estimates were within approximately 3% of calliper measurements, and the lightweight architecture is suitable for deployment on portable inspection devices, though excessive training iterations were found to cause overfitting.

2.3 Blockchain, Digital-Twin and Provenance-Management Approaches

2.3.1 PLOS ONE (2025) — "Reimagining Cultural Heritage Conservation through VR, Metaverse, and Digital Twins: An AI and Blockchain-Based Framework"

Objective: To examine how AI-driven modelling, interactive functionality, and blockchain/NFT authentication jointly affect heritage-conservation effectiveness. Methodology/Technology: A conceptual framework combining digital twins, blockchain/NFT-based authentication and immersive interaction, validated using Partial Least Squares Structural Equation Modelling (PLS-SEM) on 575 survey responses. Findings: Blockchain/NFT authentication was found to indirectly improve conservation effectiveness

through improved digital authenticity and user immersion, though the framework was validated through survey data rather than a deployed sensor-based system.

2.3.2 npj Heritage Science (2024) — "ArchaeoMeta: Leveraging Blockchain for Secure and Scalable Virtual Museums in the Metaverse"

Objective: To secure digital-artefact provenance and manage visitor interaction in metaverse-based virtual museums. Methodology/Technology: Smart contracts deployed on the Ethereum Sepolia testnet within a six-layer metaverse-museum architecture. Findings: The framework authenticated artefact provenance and handled variable visitor loads in test conditions, but scalability and cost under real mainnet deployment remain unverified.

2.4 IoT-Based Anti-Theft and Tracking

2.4.1 IEEE Access (2019) — "Study on the Anti-Theft Technology of Museum Cultural Relics Based on Internet of Things"

Objective: To overcome the unreliability of traditional infrared and door-magnetic anti-theft methods. Methodology/Technology: Passive RFID readers/writers define a safe range around each artefact; movement of a tagged artefact beyond that range triggers an immediate alarm. Findings: Hardware and software testing showed higher immediacy and a higher safety factor than conventional anti-theft methods, although passive RFID range is short and performance in crowded exhibition halls was not evaluated.

3. Comparative Analysis of Existing Approaches

Comparing the reviewed approaches on accuracy/precision, performance, scalability and efficiency highlights distinct strengths across the three technology families surveyed:

- **Accuracy/Precision:** Deep-learning crack-detection methods report high precision (e.g., ~3% error against calliper ground truth), while IoT environmental sensors offer high measurement accuracy for physical parameters (temperature, humidity, light) but do not by themselves assess structural or material condition.
- **Performance/Responsiveness:** RFID-based anti-theft systems and RF-harvesting ambience-control systems provide near-real-time alerting, whereas most blockchain-based provenance systems are optimised for tamper-resistance and auditability rather than real-time responsiveness.
- **Scalability:** Battery-less NFC/RF-harvesting sensor designs scale well for large numbers of low-power nodes but are constrained by available ambient energy and communication range; blockchain

frameworks (e.g., ERC-1155 with IPFS off-chain storage) scale well for large artefact catalogues but introduce governance and interoperability overhead across institutions.

- Efficiency: Lightweight two-stage deep-learning pipelines (classification followed by segmentation) are more computationally efficient than end-to-end instance-segmentation models, making them more suitable for edge deployment inside a museum, at some cost to detection granularity.

4. Limitations and Challenges of Existing Research

- Most IoT environmental-monitoring prototypes are validated in a single institution or case study, with limited evidence of long-term, multi-site field reliability.
- Deep-learning damage-detection models require large, accurately labelled image datasets, which are scarce and expensive to produce for diverse artefact materials and styles.
- Model performance is sensitive to lighting conditions, surface texture, and material type, reducing generalisability across different artefact categories.
- RF-energy-harvesting and NFC-powered sensor designs are constrained by available ambient energy and short communication range, limiting continuous, always-on operation at scale.
- Passive RFID anti-theft systems have limited detection range and have not been evaluated under crowded, real-world exhibition conditions.

5. Research Gap

The literature shows substantial, mature progress within each individual technology stream: IoT-based microclimate monitoring, deep-learning-based damage/crack detection, RFID-based anti-theft tracking, and blockchain-based provenance and authentication. However, no reviewed study proposes or validates a single, end-to-end Smart Museum Artifact Preservation Platform that fuses (i) real-time environmental and security sensing, (ii) AI/computer-vision-based automated condition and damage assessment, and (iii) a tamper-resistant blockchain ledger for provenance, restoration history and ownership — with all three layers exchanging data through a common decision-support dashboard. This absence of an integrated, cross-layer architecture, together with the lack of lightweight, explainable, edge-deployable AI models and standardised risk-scoring methods that translate raw sensor/vision data into actionable conservation decisions, constitutes the central research gap addressed by the assigned problem statement.

6. Possible Improvements and Future Research Directions

- Design a unified data architecture that fuses environmental sensor streams, computer-vision-based damage/condition scores, and blockchain provenance records into a single museum preservation dashboard.
- Develop lightweight, explainable deep-learning models for artefact condition and damage assessment that can run on low-power edge devices within display cases or storage rooms.
- Investigate hybrid energy-harvesting strategies (RF, NFC, solar/kinetic) to support denser, always-on sensor deployments without frequent battery replacement.
- Extend RFID/IoT anti-theft detection with computer-vision-based visitor and object-movement analytics to improve reliability in crowded exhibition environments.
- Evaluate blockchain provenance layers under mainnet-scale conditions, focusing on transaction cost, cross-chain interoperability, and integration with existing museum collection-management databases.

7. Conclusion

This literature survey reviewed twelve research works relevant to the Smart Museum Artifact Preservation Platform problem statement, spanning IoT-based environmental monitoring, deep-learning-based damage detection, RFID-based anti-theft systems, and blockchain-based provenance and authentication frameworks. While each sub-domain has produced validated, effective point solutions, the survey identifies a clear research gap: the absence of an integrated platform that combines environmental sensing, AI-based condition assessment, security tracking, and tamper-proof provenance management within a single, scalable, and field-validated system. Addressing this gap through the future research directions outlined above would enable more holistic, proactive, and trustworthy preservation of museum artefacts.

References

1. Ch.B.V. Prasanth et al., "IoT Based Environment Monitoring System To Protect Heritage Artefacts," in Proc. IEEE Conf., 2021.
2. N. A. Eltresy, O. M. Dardeer, A. Al-Habal, E. Elhariri, A. H. Hassan, A. Khattab, D. N. Elsheakh, S. A. Taie, H. Mostafa, H. A. Elsadek, and E. A. Abdallah, "RF Energy Harvesting IoT System for Museum Ambience Control with Deep Learning," *Sensors*, vol. 19, no. 20, p. 4465, 2019.
3. D. R. Gawade, S. Ziemann, S. Kumar, D. Iacopino, M. Belcastro, D. Alfieri, K. Schuhmann, M. Anders, M. Pigeon, J. Barton, B. O'Flynn, and J. L. Buckley, "A Smart Archive Box for Museum

Artifact Monitoring Using Battery-Less Temperature and Humidity Sensing," *Sensors*, vol. 21, no. 14, p. 4903, 2021.

4. "IoT-Driven Environmental Monitoring in Special Collections Preservation," *Indian Journal of Information Sources and Services*, 2025.
5. N. Wang, Q. Zhao, S. Li, X. Zhao, and P. Zhao, "Damage classification for masonry historic structures using convolutional neural networks based on still images," *Computer-Aided Civil and Infrastructure Engineering*, vol. 33, no. 12, pp. 1073-1089, 2018.
6. N. Wang, X. Zhao, L. Wang, and Z. Zou, "Novel system for rapid investigation and damage detection in cultural heritage conservation based on deep learning," *Journal of Infrastructure Systems*, vol. 25, no. 3, 2019.
7. "Deep Learning-Based Crack Detection on Cultural Heritage Surfaces," *Applied Sciences*, vol. 15, no. 14, art. 7898, 2025.
8. "HistoNet: A Hybrid Deep Learning Framework for Historical Place Image Classification," 2024.
9. "Reimagining cultural heritage conservation through VR, metaverse, and digital twins: An AI and blockchain-based framework," *PLOS ONE*, 2025.
10. "Archaeometa: leveraging blockchain for secure and scalable virtual museums in the metaverse," *npj Heritage Science*, 2024.